

Fully Online Retrieval Adaptation

Executive Summary

28 August 2026

Current evidence

No cluster experiment has yet completed in this new project. Consequently, there is no accuracy conclusion, selected ablation value, or claim that online adaptation improves a frozen backbone.

Implemented protocol

The project now precomputes T0, T1+T2, and T3 indexes before extraction. Each query retrieves from exactly $N_{\text{datastore dates}}$ mature origins, and the ridge or gate fits on exactly N_{fit} already matured dates for the query user. The defaults are 100 datastore dates, 100 fitting dates per query user, all-user raw Euclidean retrieval, same-user fitting, and a non-fixed datastore and training set. An all-user fitting scope remains available. At every forecast, the oldest 80% of the causal fitting window trains candidate ridges and the newest 20% selects $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and $K \in \{1, 5, 10, 15\}$; the chosen pair is then refitted on the complete window.

Extraction is persistent and independently manifested, allowing changes to adaptation parameters to reuse matching features. Ridge writes rolling coefficient means and standard deviations; CatBoost writes rolling feature importance means and standard deviations. Ridge also records each per-query selected pair and validation loss. The fixed-protocol ablation crosses explicit fixed-datastore and fixed-training-set booleans while retaining one identical T3 query grid. Reports reject date mismatches and average models only across the same dataset and $L:H$ configurations.

Planned evidence

The first required evidence is the Electricity test profile from `slurm/dgx/main/01_main_online_ridge.slurm`. It must establish that checkpoint loading, causal extraction, full-capacity datastore limits, the matched ridge, convex, Bayes, direct-covariate, and vanilla rows, identical-date reporting, and cache size are operational. The second smoke run is `slurm/dgx/main/02_tsrag_comparison.slurm`. It must validate the native TS-RAG fixed same-variable training datastore, $K = 5$, and the shared 512:64 evaluation grid.

Only after these checks should the small profile be run on Electricity, Solar, and Traffic. The full profile then adds raw ETT and prepared TIME datasets. The resulting ablations will characterize alpha/K sensitivity without using final evaluation outcomes for selection, determine whether $N_{\text{datastore dates}}$ and N_{fit} should differ, and how retrieval and fitting scope interact across users.

Interpretation limit

The Chronos-Bolt SOTA table places causal online MSE beside published TS-RAG and Cross-RAG MSE only as contextual evidence. The dedicated TS-RAG workflow does reproduce its fixed same-variable retrieval scope and split boundaries, but preprocessing and metric-scale differences still have to be checked before combining local and published numbers.